LIVE WEBSITE QUALITY REVIEW

CivicLens QA report

Verdict: The map is usable, but the deployed site is not ready for an unscripted demo of the simulator and planning assistant. Fix the simulator crash, missing evidence index, and contradictory area statistics first.

Tested: https://civiclens-sotonship2026.onrender.com/
18 October 2026 | Desktop Chrome | Observed viewport: 1363 x 936

ID / priority	Confirmed finding	Impact
F01 / Critical	Simulator crashes from both entry points.	Core workflow unavailable.
F02 / High	Two typed assistant questions return a missing-index error.	Free-text analysis fails.
F03 / High	Area comparisons display Infinity multipliers.	Invalid numeric evidence.
F04 / High	Underserved-sector section shows false zero counts.	Misleading opportunity signals.
F05 / Medium	Healthy evidence indicators remain after retrieval fails.	Readiness is overstated.
F06 / Medium	Postcodes without the expected internal space return no matches.	Valid searches fail.
F07 / Low	No-result message quotes an empty detailed-type query.	Unhelpful recovery message.
F08 / Medium	Assistant uses 2023 vacancy figures without a year.	Older data can look current.
F09 / Medium	"Leading sector" changes definition with map mode.	Answers lack metric context.

Fix order: F01 > F02/F05 > F03/F04 > F08/F09 > F06/F07. Priorities are QA judgments; findings may share underlying causes. Suggested fixes below are diagnostic directions, not verified source-code diagnoses.

01 / AVAILABILITY

Restore the core workflows

F01 | Critical | Simulator fails during initialization

Reproduce: From the landing page, click Explore the map, then Simulator. Separately, recover with Back to the start and click Test a site.

Actual: Both routes display the full-page error below, before a scenario can be configured. The app navigation disappears. Back to the start successfully returns to the landing page.

```
Cannot read properties of undefined (reading 'create')
```

Evidence: The browser console identifies SimulatorView-B7koqUtR.js at line 5, column 1469, inside a useMemo callback. Asset path: /assets/SimulatorView-B7koqUtR.js.

Fix direction: Identify the undefined dependency or object used by that create call; check the production import/export and initialization path. Keep the error boundary, but retain usable navigation and a friendly recovery message.

Acceptance: Both entry points and a direct simulator refresh render the input controls. A valid scenario completes. No create-related exception occurs.

F02 | High | Typed questions fail with missing evidence index

Reproduce: Open Ask the assistant and submit: "What planning restrictions should I investigate before opening a new cafe in Portswood?"

Actual: The response is "Index not found. Run `sotonship-rag index` first." An out-of-scope Tokyo question also produced this error. Suggested questions about business counts and vacancy did return answers; the comparison shortcut also worked after errors.

Fix direction: Build and deploy the retrieval index and verify the runtime can find it. Check index path and persistence in the deployed environment. Replace internal setup instructions with a user-facing unavailable message.

Acceptance: The exact planning question returns supported evidence or a clear evidence limitation. Unsupported questions are handled gracefully. Successful preset answers alone do not pass this check.

F05 | Medium | Evidence status does not reflect retrieval readiness

Reproduce / actual: After F02, the header still says Evidence service online and the assistant says Evidence connected. This may report service reachability, but it does not signal that retrieval is unavailable.

Fix / acceptance: Distinguish reachable, index ready, and degraded states. Check retrieval readiness before showing a healthy evidence indicator; show a degraded state when the index cannot be used.

Repair the area comparisons

F03 | High | Non-finite business concentration ratios

Reproduce: Explore the map > Popular > Stands out vs city. Click Portswood on the map, then inspect What's here. Switch to Bargate using Other areas.

Actual: All eight displayed business-type comparisons in each tested area show Infinity x. Examples: Portswood Cafe, 24 places; Bargate Cafe, 78 places.

Expected: Finite ratios when citywide reference counts are nonzero. The site itself reports 221 cafes in 3,718 places, with 255 places in Portswood and 1,060 in Bargate.

Check using the displayed definition	Expected approximate ratio
Portswood: (24 / 255) / (221 / 3,718)	1.58 x
Bargate: (78 / 1,060) / (221 / 3,718)	1.24 x

Fix direction: Check that the area numerator and city denominator use the same detailed-type keys, population of places, and filter state. Handle zero/missing denominators explicitly. Do not substitute an invented finite value.

Acceptance: These examples match the displayed definition after rounding. No Infinity or NaN reaches the interface. A genuinely undefined ratio is labelled unavailable with a reason.

F04 | High | Opportunity section contradicts observed sector counts

Reproduce: In the same unfiltered Portswood area profile, inspect Less of than the city average and compare it with the map tooltip for Portswood.

Actual: The section lists Food and drink - 0 and Retail - 0. The area tooltip shows Food and drink 90 and Retail 71. The area total is 255.

Why it matters: Food and drink represents about 35.3% of Portswood places versus 29.0% citywide (1,079 / 3,718). Under the displayed share-based comparison, it should not be presented as a zero-supply gap. A planner could mistake this for a real unmet need.

Fix direction: Verify broad-sector and detailed-type identifiers are not mixed. Calculate sector counts and comparisons from one consistent aggregation. A missing lookup must not silently become zero.

Acceptance: Counts agree between tooltips, filters, and area profiles. The underserved list follows a clearly stated metric and excludes categories that do not meet it.

Important: F03 and F04 may share an aggregation defect, but this cannot be established without reviewing the code. The discrepancies themselves were observed directly.

03 / ASSISTANT CORRECTNESS

Make the evidence unambiguous

F08 | Medium | Vacancy answer omits the year and uses older figures

Reproduce: Ask the assistant using Compare Shirley and Portswood. Open its Southampton Business Opportunity Data, p. 6 citation.

Actual: The assistant reports Shirley 10.94% and Portswood 6.45%, without a year. Page 6 labels those figures Rate 2023 and also provides Rate 2024 values of 10.57% and 4.84%, respectively. The newer figures are not mentioned.

Centre	2023 cited by assistant	2024 also in source
Shirley	10.94%	10.57%
Portswood	6.45%	4.84%

Expected: Use the most recent applicable column by default, or explicitly label a historical comparison. Every vacancy statistic should carry its observation year. This does not establish actual vacancy in 2026.

Fix / acceptance: Store the observation year with each value and bind answer generation to the selected period. The comparison should say which year it uses and match that exact table column.

Verified source: /api/evidence/data/raw/project/southampton_business_opportunity_data.pdf#page=6. The PDF opened successfully at page 6; it describes an October-December 2024 unit survey.

F09 | Medium | "Leading sector" silently changes meaning with mode

Reproduce: With Popular set to Stands out vs city, ask Compare Shirley and Portswood. Close the assistant, select Most common, reopen it and ask the same comparison.

Actual: In Stands out vs city, the answer calls Health the leading sector for both areas. In Most common, it calls Food and drink the leading sector for Portswood and Retail for Shirley. Both use the same unqualified wording.

Expected: The answer should distinguish the largest sector by count from the most overrepresented sector relative to the city. A change in map mode can change the metric, but that change must be stated.

Fix / acceptance: Pass metric names and mode context with the statistics. Say "largest by number of mapped places" or "most overrepresented relative to the city" as appropriate. The same question should either retain a stable definition or explicitly explain the active one.

The business-count shortcut matched the displayed area totals for Bargate and Portswood in the sampled checks. This is a consistency check, not independent verification of the underlying dataset.

Polish search, then retest

F06 | Medium | Postcode spacing causes false no-results

Reproduce / actual: In Places, with all industries and no area/type filter, search SO14 2BY: 14 places. Lowercase so14 2by also returns 14. SO142BY or SO14 2BY (two internal spaces) returns 0.

Fix / acceptance: Normalize postcode whitespace and case for postcode matching. All four representations should return the same results while ordinary name/street search continues to work.

F07 | Low | No-results copy refers to an empty type query

Reproduce / actual: Leave Detailed type empty and search zzzz_QA_no_such_place_123 in Name, postcode, street. The UI shows 0 places and No type matches "".

Fix / acceptance: Use a venue-search empty state that includes the actual query and offers to clear filters. Reserve type-specific copy for searches in the Detailed type field.

Checks that worked

Area	Observed result
Map controls	Popular/Heatmap/People/Places switch; sampled sector and age filters update counts.
Place exploration	Area drill-down, area switching, type drill-down and clear controls respond. Tesco marker opens address, postcode, source and confidence.
Search robustness	Tesco, tesco and padded Tesco return 32. Harmless markup/punctuation and 2,048 characters produce zero results without crashing.
Filter consistency	Health gives 122; Pharmacy gives 42; changing to Food and drink clears the incompatible subtype and gives 1,079.
Assistant / sources	Whitespace-only send is disabled. Sampled shortcuts respond; source disclosure expands and the vacancy PDF opens at page 6.
Recovery	Back to the start recovers from the simulator crash. Assistant can still answer a shortcut after the missing-index error.

Scope and limits: Live UI and browser-console observations only. No repository, backend logs, database, or deployment was changed. Mobile layouts, other browsers, full keyboard/screen-reader accessibility, load/concurrency, server-side validation and security were not tested. The crash blocked simulator-input, scenario-output and export testing. Search robustness does not establish XSS or SQL-injection safety.

Next release gate: Retest the exact failures above, then complete one end-to-end scenario, one arbitrary evidence question, a mobile layout pass and a second-browser smoke test.